

LVA-3 Migration to canonical workable-items — EXECUTION LOG (COMPLETE + VERIFIED)

Executed: 2026-05-31 by execution subagent (LVA-3, per docs/tickets/MIGRATION-T0-CANONICAL.md rev 1). **Outcome:** COMPLETE. Lava ticket tracking migrated from the bespoke tools/lava-tickets/ system to the **canonical** workable-items Go binary (constitution/scripts/workable-items/), keyed LVA-N. All 7 LVA keys preserved verbatim. New gate CM-WORKABLE-ITEMS-SYNC passes; falsifiability rehearsed. **§11.4.6 (no-guessing):** every result below is real captured command output. Two early missteps are recorded honestly: (a) a first seed pass used placeholder titles/bodies and was DISCARDED + rebuilt with verbatim real data; (b) the first body_md status-patch used the wrong Status-line literal, which the finisher fixed via auto-detection of the generator's exact prefix. **NO git commit/push performed** — the main agent stages with `git add -A` (exact list in §9).

§1 — Build the canonical binary (VERIFIED)

```
( cd constitution/scripts/workable-items && CGO_ENABLED=1
GOMAXPROCS=2 nice -n 19 go build -o bin/workable-items ./cmd/
workable-items )
```

→ **EXIT 0**, binary = **7,437,170 bytes**. (bin/ is gitignored in the submodule — not a tracked change.) `CGO_ENABLED=1` is required; `root go build` fails (no `root go.mod`).

§2 — Read the old DB (VERIFIED — 7 rows, NOT 8)

sqlite3 docs/tickets/tickets.db "SELECT COUNT(*) FROM tickets;" → 7. No LVA-8 exists in this checkout. Verbatim real rows captured (titles/bodies/provenance):

id	LVA type	LVA status/ closure	prio	title (verbatim)
LVA-1	Bug	Closed/Fixed	P1	Deflake CredentialsViewModelTest > select provider updates selectedProvider
LVA-2	Debt	Closed/Closed	P1	§6.X-debt darwin/arm64 emulator-acceleration sub-debt
LVA-3	Task	InProgress	P1	Adopt HelixConstitution §11.4.79–106 (pin 53 commits behind)
LVA-4	Feature	InProgress	P1	LVA workable-items ticket system (SQLite DB + Go CLI + md/HTML/PDF/ DOCX export)
LVA-5	Incident	OperatorBlocked	P0	Rotate Firebase CI token (printed to session transcript)
LVA-6	Task	Open	P2	§11.4.79 reconcile codegraph index policy (own-org submodules IN index)
LVA-7	Task	Open	P2	§11.4.85 stress + chaos test scaffold

Real LVA-5 operator_blocked_details (verbatim): "Operator must run firebase logout then firebase login:ci and update LAVA_FIREBASE_TOKEN in gitignored .env. Agent cannot rotate credentials."

§3 — Seed canonical DB + statuses (VERIFIED)

`rm -f docs/workable_items.db{-wal,-shm}, 7× add --id LVA-N (all rc 0), type-mapping per plan §3 (LVA-2 Debt→Task, LVA-5 Incident→Bug; severity = P-level). Closes via close; non-terminal statuses via DB-direct UPDATE (no update subcommand). The repeatable seed+verify script is committed at docs/tickets/finish-migration.sh (idempotent — rebuilds the DB from scratch, auto-detects the generator's Status-line literal, regenerates, validates, diffs, runs the gate+test).`

FINAL canonical state (matches plan §3 target table EXACTLY):

LVA-1	Bug	Fixed (→ Fixed.md)	P1	Fixed
LVA-2	Task	Completed (→ Fixed.md)	P1	Fixed
LVA-3	Task	In progress	P1	Issues
LVA-4	Feature	In progress	P1	Issues
LVA-5	Bug	Operator-blocked	P0	Issues
LVA-6	Task	Queued	P2	Issues
LVA-7	Task	Queued	P2	Issues

LVA-N keys stored verbatim (no zero-padding) — VERIFIED. Closes recorded real evidence paths:

- LVA-1 → feature/credentials/src/test/kotlin/lava/feature/credentials/CredentialsViewModelTest.kt
- LVA-2 → .lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json

LVA-5 operator_block_details row inserted (what / why-exhausted / unblock-condition / who=Operator), reflecting the Firebase-token-rotation operator-block.

§4 — Generate docs + validate + diff (VERIFIED)

```
sync db-to-md --out-issues docs/Issues.md --out-fixed docs/
Fixed.md → wrote docs/Issues.md + docs/Fixed.md
validate
→ validate: OK — 7 items, all invariants satisfied
diff --issues docs/Issues.md --fixed docs/
Fixed.md → diff: DB and Markdown are in sync
[exit 0]
report --by-
status → In
progress 2 | Queued 2 | Completed 1 | Fixed 1 | Operator-blocked 1
```

```
| TOTAL 7
PRAGMA
wal_checkpoint(TRUNCATE) →
WAL collapsed
```

Output paths CONFIRMED from the binary: docs/Issues.md (LVA-3/4/5/6/7) + docs/Fixed.md (LVA-1/2) at the docs ROOT. The binary does NOT emit _Summary.md/HTML/PDF/DOCX (those were tools/lava-tickets/export.go/render.go only — see §7 OWED). A stale, INCORRECT docs/Issues_Summary.md (+Fixed_Summary.md) leftover from an earlier generation (wrong data: LVA-4 Task vs Feature, LVA-5 P1 vs P0) was REMOVED so no stale summary ships (plan §3 Step 3).

§5 — Gate replacement (CM-LVA-TICKETS-SYNC → CM-WORKABLE-ITEMS-SYNC) (VERIFIED)

grep -rl located the wiring (the plan's UNCONFIRMED location):

- gate: scripts/check-lva-tickets.sh
- sweep: scripts/verify-all-constitution-rules.sh **lines 189–193**
(run_gate "lva-tickets-sync" ... "bash scripts/check-lva-tickets.sh --advisory")
- test: tests/check-lva-tickets/test_check_lva_tickets.sh (+ EVIDENCE.md)
- gates-index: docs/helix-constitution-gates.md line 25
- script docs: docs/scripts/check-lva-tickets.sh.md, docs/scripts/lava-tickets.md
- (no .github/hooks/pre-push reference)

Actions:

- **Created** scripts/check-workable-items.sh (chmod +x) — canonical validate + diff (exit 1 on divergence) + git ls-files --error-unmatch docs/workable_items.db (§11.4.95). Builds binary if absent.
- **Created** tests/check-workable-items/test_workable_items_sync.sh (chmod +x) — hermetic falsifiability test.
- **Created** docs/scripts/check-workable-items.sh.md (§11.4.18 script doc — keeps CM-SCRIPT-DOCS-SYNC green).
- **Edited** scripts/verify-all-constitution-rules.sh lines 189–193 →
run_gate "workable-items-sync" "CM-WORKABLE-ITEMS-SYNC ..."
"bash scripts/check-workable-items.sh" (strict; DB+docs ship in-sync).

- **Edited** docs/helix-constitution-gates.md line 25 → new CM-WORKABLE-ITEMS-SYNC row.
- **Deleted** old gate + test + EVIDENCE + the 2 stale script docs.

§6 — Gate run + falsifiability rehearsal (VERIFIED, real evidence)

Reversible probe (git add -N docs/workable_items.db ... run ... git reset; NO commit, DB content untouched):

```
GATE: CM-WORKABLE-ITEMS-SYNC: OK — docs/workable_items.db
validated, DB ↔ Markdown in sync, DB tracked [gate_rc=0]
TEST: positive case: PASS
      negative case (corrupted docs/Issues.md): gate correctly
FAILED
      (restore) → PASS
      PASS: CM-WORKABLE-ITEMS-SYNC gate is
falsifiable
[test_rc=0]
```

Pre-staging (DB not yet tracked) the gate correctly FAILS with the §11.4.95 directive — proving the tracked-check works; it goes green once the main agent stages the DB. scripts/check-script-docs-sync.sh → ✓ all 71 scripts documented; no orphan docs (EXIT 0) after the doc add + stale-doc removals.

§7 — Retire bespoke system (VERIFIED deleted via plain rm)

- tools/lava-tickets/ (whole dir: _seed_evidence.sh, commands.go, export.go, go.mod, go.sum, main.go, render.go, tickets_test.go; bin/ was gitignored).
- docs/tickets/tickets.db; docs/tickets/{Issues,Fixed,Issues_Summary,Fixed_Summary}.md; docs/tickets/export/ (docx + html subtrees).
- scripts/check-lva-tickets.sh; tests/check-lva-tickets/ (test + EVIDENCE.md); docs/scripts/check-lva-tickets.sh.md; docs/scripts/lava-tickets.md.
- Stray docs/Issues_Summary.md + docs/Fixed_Summary.md (non-canonical leftovers).

KEPT under docs/tickets/ as historical record (per dispatch + plan §4 Step 4): DESIGN.md, BUILD-EVIDENCE.md, MIGRATION-TO-CANONICAL.md, schema.sql, this MIGRATION-EXECUTED.md, and finish-migration.sh (the repeatable seeder).

Minor plan discrepancy: plan §4 Step 4 says delete docs/tickets/schema.sql; the dispatch said KEEP it. Honored the dispatch (KEPT). Main agent may delete if preferred — no longer load-bearing.

OWED follow-ups (NOT blocking; file as future LVA items):

- Canonical binary has no update/reopen/block subcommand — non-terminal/operator-blocked transitions used DB-direct UPDATE. Upstream-extend per §11.4.74 (constitution-submodule §11.4.26 7-step pipeline).
- Summaries (*_Summary.md) + HTML/PDF/DOCX exports are NOT emitted by the canonical binary. If still required, regenerate via the project export pipeline / §11.4.106 Docs Chain.
- Incident/Debt/Chore→Task coercion caveat + source/source_ref folded into description (documented in CLAUDE.md §6.AF + CONTINUATION by the main agent).
- docs/chaos-stress/EXPORT-AUDIT.md references the old lava-tickets export pipeline — main agent should update or mark historical.

§8 — .gitignore (VERIFIED)

Replaced the dead # lava-tickets build artifacts block with:

```
# workable-items canonical tracker – DB is TRACKED (§11.4.95);  
only WAL/SHM sidecars are ignored  
docs/workable_items.db-wal  
docs/workable_items.db-shm
```

Verified: git check-ignore docs/workable_items.db-{wal,shm} → rc 0 (both ignored, correct); git check-ignore docs/workable_items.db → rc 1 / empty (**NOT ignored = correct, §11.4.95**). WAL/SHM sidecars checkpoint-collapsed + removed for a clean tree.

§9 — Precise staging list for the main agent (git add -A captures all of this)

CREATE (untracked ??):

- docs/workable_items.db ← §11.4.95 TRACKED — must be staged
- docs/Issues.md
- docs/Fixed.md
- scripts/check-workable-items.sh
- tests/check-workable-items/test_workable_items_sync.sh
- docs/scripts/check-workable-items.sh.md
- docs/tickets/MIGRATION-EXECUTED.md (this file)
- docs/tickets/finish-migration.sh (repeatable seeder/verifier)

MODIFY (M):

- .gitignore
- docs/helix-constitution-gates.md
- scripts/verify-all-constitution-rules.sh

DELETE (D):

- scripts/check-lva-tickets.sh
- tests/check-lva-tickets/test_check_lva_tickets.sh, tests/check-lva-tickets/EVIDENCE.md
- docs/scripts/check-lva-tickets.sh.md, docs/scripts/lava-tickets.md
- docs/tickets/tickets.db
- docs/tickets/Issues.md, docs/tickets/Fixed.md, docs/tickets/Issues_Summary.md, docs/tickets/Fixed_Summary.md
- docs/tickets/export/docx/{Issues,Fixed,Issues_Summary,Fixed_Summary}.docx
- docs/tickets/export/html/{Issues,Fixed,Issues_Summary,Fixed_Summary}.html
- tools/lava-tickets/{_seed_evidence.sh,commands.go,export.go,go.mod,go.sum,main.go,render.go,tickets_test.go}

KEPT untouched: docs/tickets/{DESIGN,BUILD-EVIDENCE,MIGRATION-TO-CANONICAL}.md, docs/tickets/schema.sql. **WAL/SHM sidecars:** removed; gitignored regardless.

Note on `git add -A` vs the stray summaries: `docs/Issues_Summary.md` + `docs/Fixed_Summary.md` at the docs ROOT were untracked stray files; they were `rm'd`, so they will NOT be staged (they never were tracked).

§10 — §6.S / §6.AF follow-ups OWED (main agent — no commit allowed in this subagent)

- CLAUDE.md §6.AF: rewrite to point at the canonical binary (`constitution/scripts/workable-items/`), DB `docs/workable_items.db`, docs `docs/Issues.md/docs/Fixed.md`, the `--id LVA-N (preserve) / --prefix LVA (new)` convention, the Incident→Bug / Debt→Task / Chore→Task mapping, and the new CM-WORKABLE-ITEMS-SYNC gate. Add Catalogue-Check: reuse `HelixDevelopment/HelixConstitution@<pin-sha> scripts/workable-items + Classification: project-specific`.
- `docs/CONTINUATION.md`: LVA-3 done; LVA-4 superseded-by-migration; new DB+doc paths; retired `tools/lava-tickets/`; the OWED follow-ups from §7.